

PRACTICAL – 5

Registration Form with Frontend Validation and User-Friendly Error Handling

AIM:

To build the StudentHub registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender and terms acceptance, using regular expressions to check each field and a password strength meter, while keeping the form accessible with the keyboard and with screen readers.

Introduction:

The registration form on register.html was extended with the fields required by the sheet and a new script, js/validation.js, that checks each field as the user types and again when the form is submitted. The browser's default validation pop-ups were turned off with the novalidate attribute so that custom error messages could be shown next to each field instead.

Tools and Technology:

VS Code, JavaScript (ES5/ES6), HTML5, CSS3, regular expressions, browser developer tools (Console), Git and GitHub.

Validation Rules Used:

Field	Rule	Regular expression / check
Full name	2–50 letters, spaces, dot, apostrophe or hyphen	/^[A-Za-z][A-Za-z .'']{1,49}\$/
Email	Standard text@domain.tld shape	/^[^\s@]+@[^\s@]+\.[A-Za-z]{2,}\$/
Mobile number	10 digits, starting with 6, 7, 8 or 9	/^[6-9][0-9]{9}\$/
Password	At least 8 characters, with uppercase, lowercase, a number and a symbol	checked with /[A-Z]/, /[a-z]/, /[0-9]/ and /[^A-Za-z0-9s]/
Confirm password	Must match the password field exactly	string comparison
Course / Year / Gender	Must be selected, none left blank	required check
CAPTCHA	Typed text must match the code shown on the canvas	string comparison, not case sensitive
Terms	Checkbox must be ticked	required check

Every field has its own error message shown directly beneath it, and the field is marked with aria-invalid so screen readers announce the problem. Validation runs on keyup, input, change and blur so mistakes are caught immediately, and again on submit so nothing can be skipped.

Screenshots:

The empty form, each invalid case, and a fully valid submission were tested in the browser.

NEW ACCOUNT
+ Create account

Fill in your details to get started with StudentHub.

Please be 10 years below and try again.

Full Name

Enter your full name

Please enter your full name.

Email Address

email@example.com

Please enter your email address.

Mobile Number

01234 567890

10 digits, starting with 0, 7, 8 or 9.

Please enter your mobile number.

Password

Create a password

At least 8 characters with an uppercase letter, a lowercase letter, a number and a symbol.

Please create a password.

Confirm Password

Re-enter your password

Please re-enter your password.

Course

Select your course

Please select your course.

Year

Select your year

Please select your year.

Gender

Male Female Other

Please select your gender.

I am a

Student

Security check

g T P T F New code

Type the 5 characters from the image. Not case sensitive.

Please type the characters shown in the image.

I accept the terms and conditions

You must accept the terms and conditions to register.

Create StudentHub Account

Already have an account? Login here

Figure 1: Empty submission showing every required-field error

NEW ACCOUNT
+ Create account

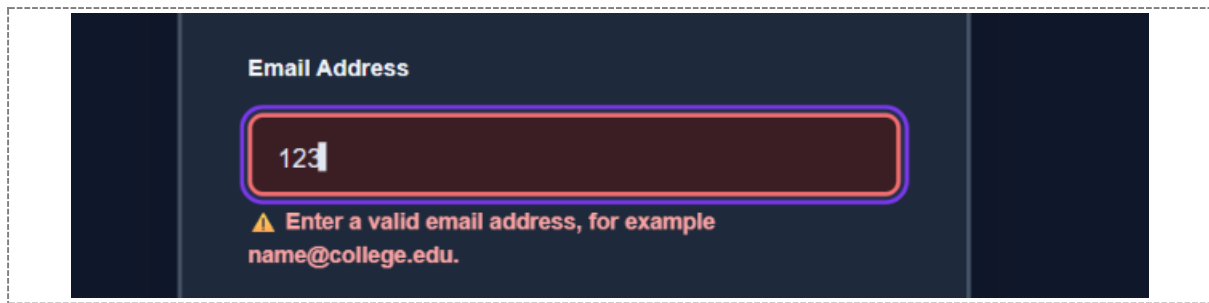
Fill in your details to get started with StudentHub.

Full Name

123

Use 2 to 50 letters. Spaces, dots, apostrophes and hyphens are allowed; numbers are not.

Figure 2: Name field rejecting digits



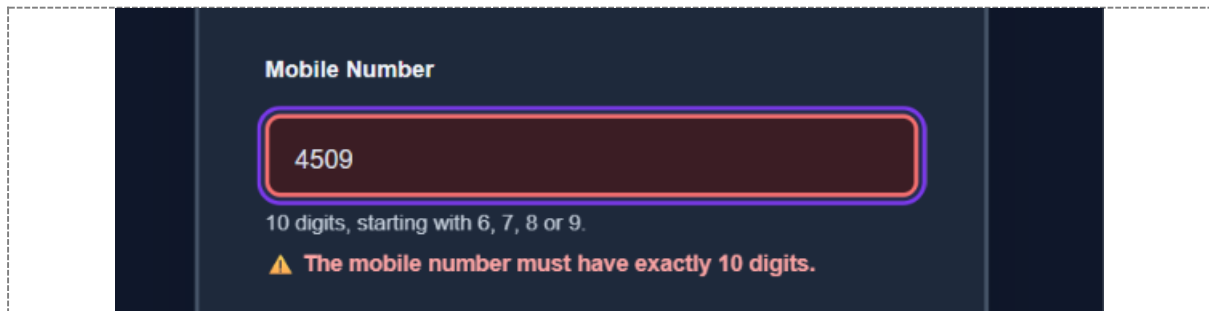
Email Address

123

⚠ Enter a valid email address, for example name@college.edu.

This figure shows a dark-themed form with a label 'Email Address' above a text input field. The input field contains the text '123'. Below the input field, there is a red warning icon followed by the text 'Enter a valid email address, for example name@college.edu.'.

Figure 3: Email field rejecting a bad address



Mobile Number

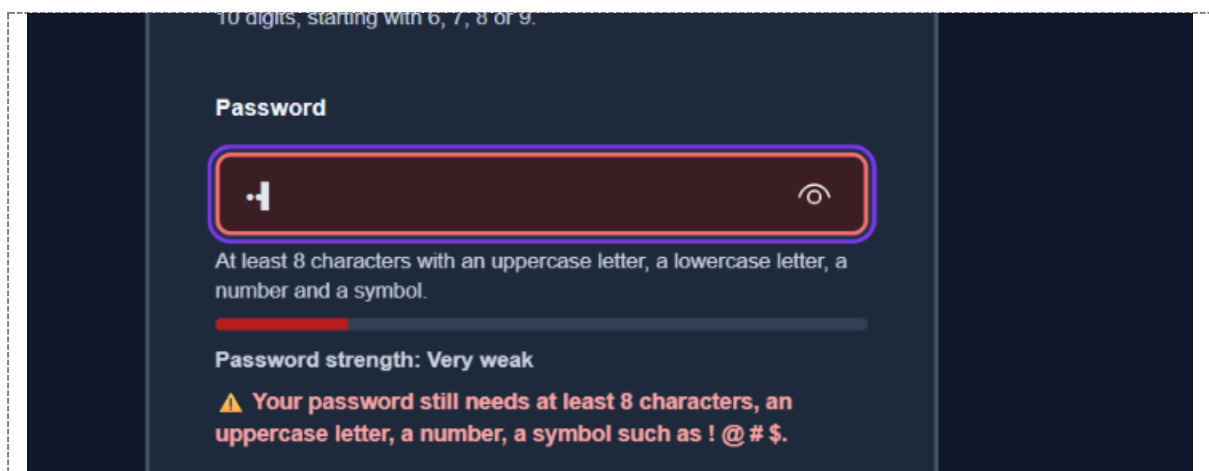
4509

10 digits, starting with 6, 7, 8 or 9.

⚠ The mobile number must have exactly 10 digits.

This figure shows a dark-themed form with a label 'Mobile Number' above a text input field. The input field contains the text '4509'. Below the input field, there is a red warning icon followed by the text 'The mobile number must have exactly 10 digits.'.

Figure 4: Mobile number validation



Password

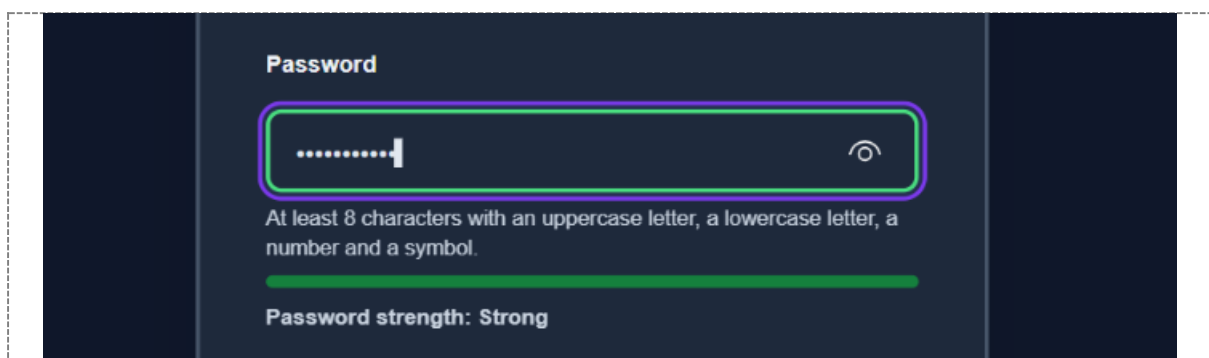
At least 8 characters with an uppercase letter, a lowercase letter, a number and a symbol.

Password strength: Very weak

⚠ Your password still needs at least 8 characters, an uppercase letter, a number, a symbol such as ! @ # \$.

This figure shows a dark-themed form with a label 'Password' above a text input field. The input field contains a weak password, represented by a few characters and a red eye icon. Below the input field, there is a red warning icon followed by the text 'Your password still needs at least 8 characters, an uppercase letter, a number, a symbol such as ! @ # \$.'.

Figure 5(a): Password strength meter on a weak password



Password

At least 8 characters with an uppercase letter, a lowercase letter, a number and a symbol.

Password strength: Strong

This figure shows a dark-themed form with a label 'Password' above a text input field. The input field contains a strong password, represented by dots and a red eye icon. Below the input field, there is a green bar indicating a strong password.

Figure 5(b): Password strength meter on a strong password

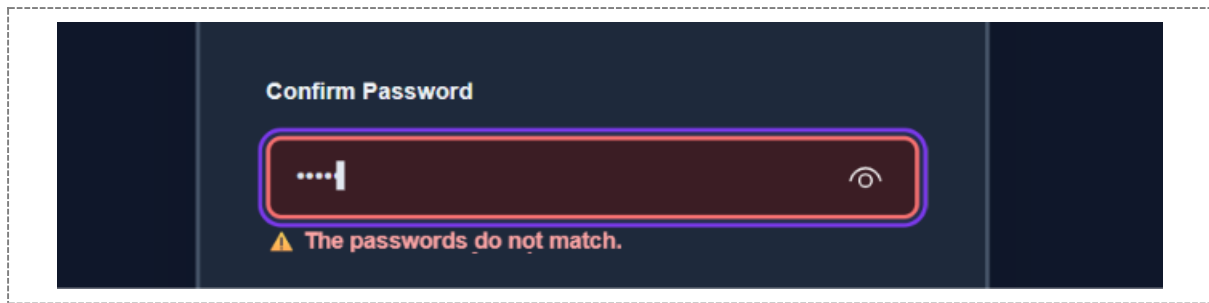


Figure 6: Confirm-password mismatch

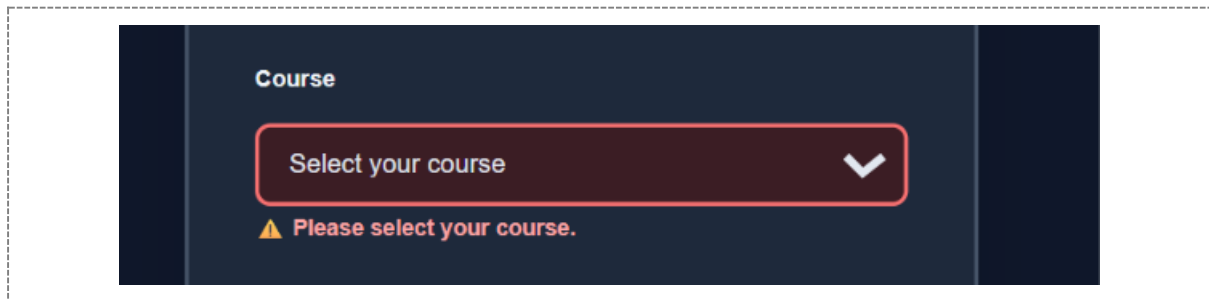


Figure 7: Course and year required errors

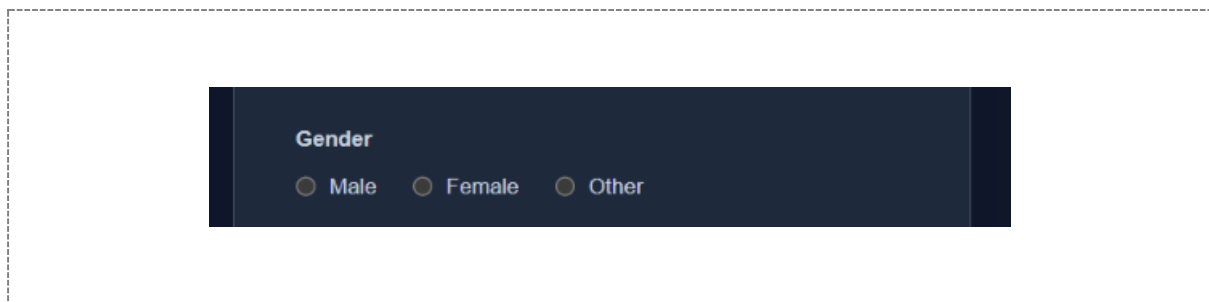


Figure 8: Gender required error

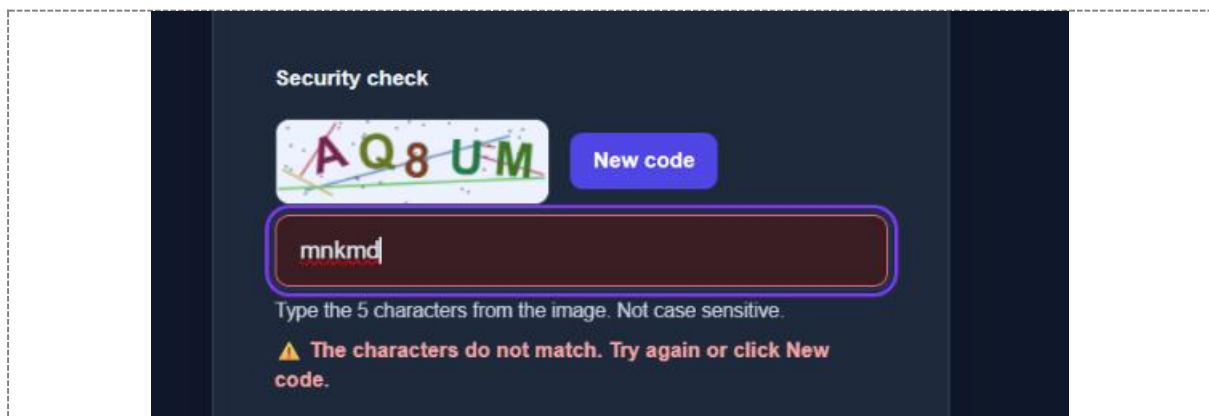


Figure 9: CAPTCHA mismatch

NEW ACCOUNT
Create account

Fill in your details to get started with BlunderHub.

Full Name
Daniel Rupert

Email Address
danielrupert123@gmail.com

Mobile Number
9192109180
(11 Digits, starting with 6, 7, 8 or 9)

Password
xxxxxxxxxx
At least 8 characters with an uppercase letter, a lowercase letter, a number and a special

Password strength: Strong

Confirm Password
xxxxxxxxxx

Course
B.Tech Computer Engineering

Year
Second year

Gender
☒ Male ☐ Female ☐ Other

I am a
Student

Security check

Type the 5 characters from the image. Not case sensitive.

☒ I accept the terms and conditions

Create BlunderHub Account

ALREADY HAVE AN ACCOUNT? [Login here](#)

Figure 10: Terms and conditions required error

NEW ACCOUNT!
Create account

Fill in your details to get started with ShantitHub.

Full Name
ShantitHub

Email Address
shantitHub123@gmail.com

Mobile Number
9192109180
(11 Digits, starting with 6, 7, 8 or 9)

Password
password
At least 8 characters with an uppercase letter, a lowercase letter, a number and a special

Confirm Password
password

Course
B.Tech Computer Engineering

Year
Second year

Gender
☒ Male ☐ Female ☐ Other

Role
Student

Security check

Type the 5 characters from the image. Not case sensitive.

☒ I accept the terms and conditions

Create ShantitHub Account

ALREADY HAVE AN ACCOUNT? [Log in here](#)

Figure 11: Successful registration after passing all checks

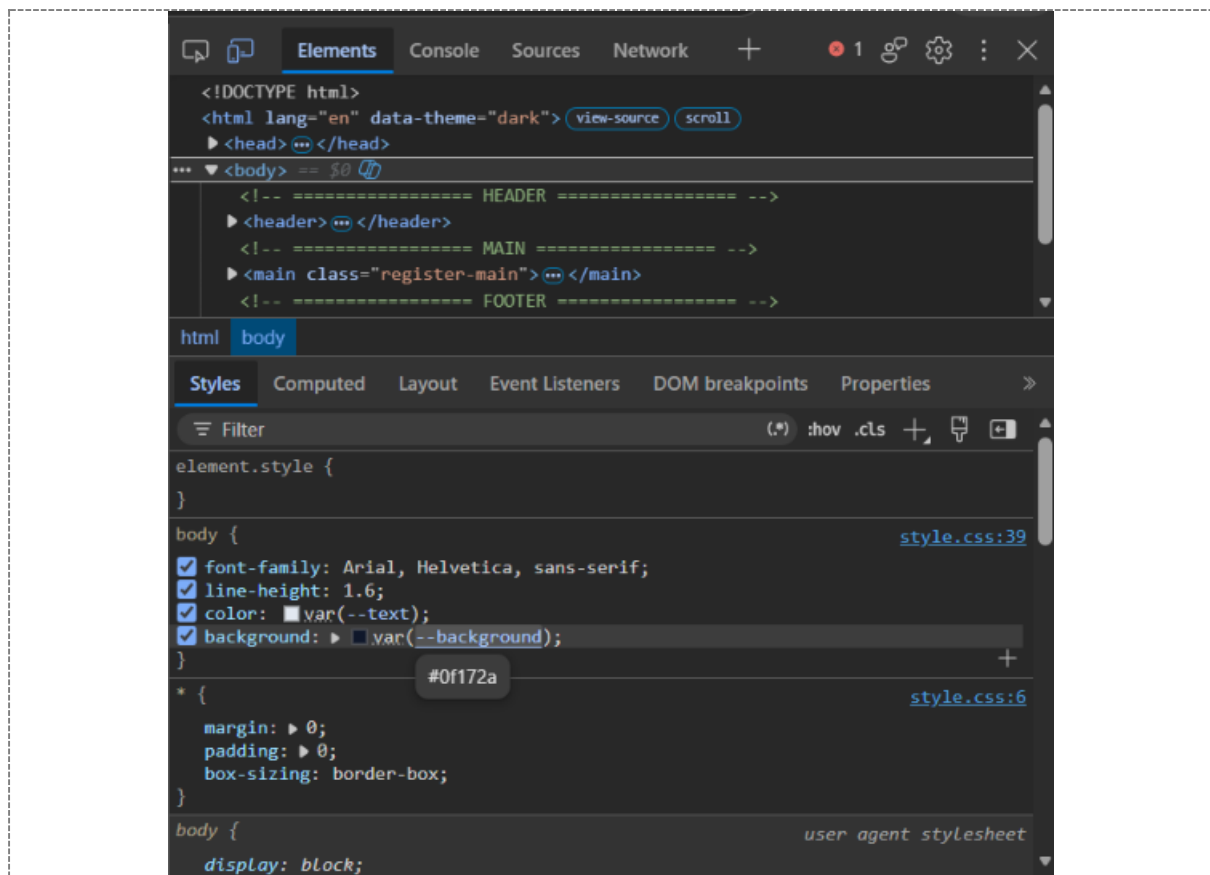


Figure 12: Console during testing

Key Questions and Answers

Q1. Are correct input types and attributes used?

Yes. The form uses `type="email"` for email, `type="tel"` for mobile, `type="password"` for the password fields, and radio buttons and a checkbox for gender and terms. `required`, `minlength`, `maxlength`, `inputmode` and `autocomplete` are set on the relevant fields, and `novalidate` on the form lets the custom JavaScript messages appear instead of the browser's default pop-ups.

Q2. Are validation errors displayed near the relevant fields?

Yes. Every field has its own `<p class="error-message">` placed directly below it, linked to the field with `aria-describedby`. This keeps each message next to the field it belongs to rather than in one combined list, so it is clear which field needs fixing.

Q3. Is password strength checked?

Yes. Five checks are run on the password (length of at least 8, an uppercase letter, a lowercase letter, a number and a symbol), and a meter fills in proportionally with a label from Very weak to Strong. The same checks decide the password error message, listing exactly what is still missing.

Q4. Is the form accessible using keyboard and screen-reader-friendly labels?

Yes. Every input has a `<label for>` pointing to its id, so clicking a label focuses its field and screen readers announce the label with the field. All controls are native elements reachable with Tab. Invalid fields get `aria-invalid="true"`, and the status message after submitting uses `role="status"` so it is announced automatically.

Post Laboratory Work:

Validation test cases with screenshots for valid and invalid inputs were captured and are included above.

GITHUB REPOSITORY LINK:

<https://github.com/DarshilRupani/studenthub-portal>

(FOR SOURCE CODE)

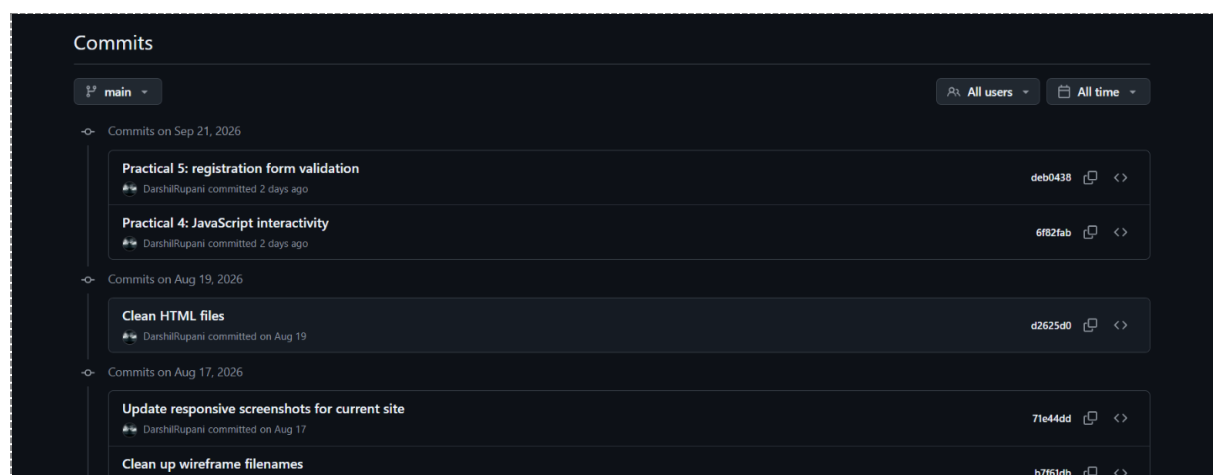


Figure 13: Practical 5 commit on GitHub

Conclusion:

The StudentHub registration form was extended with name, email, mobile, password, confirm password, course, year, gender, a CAPTCHA and terms acceptance, validated using regular expressions and JavaScript. A password strength meter, real-time validation and a canvas-based CAPTCHA were

added as extensions, and error messages were placed next to each field with ARIA attributes to keep the form accessible.